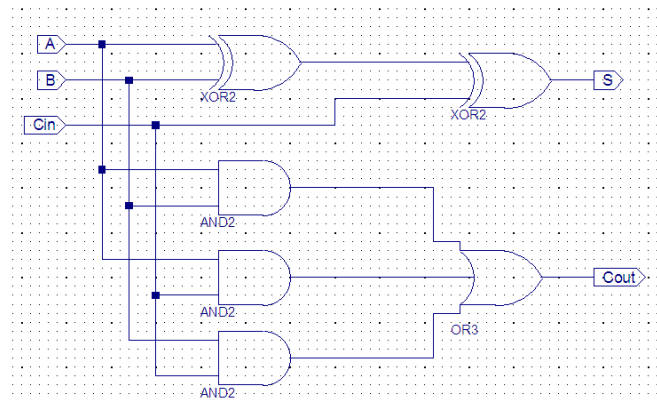


# EE533 Lab2

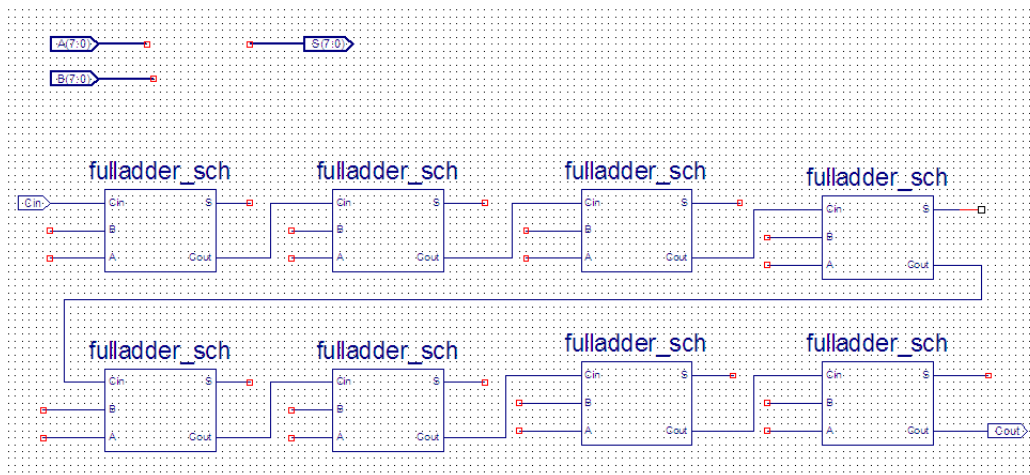
I-Hsuan Lin 5966228225

## 1. Designing and Simulating a Synchronous 8-bit Adder

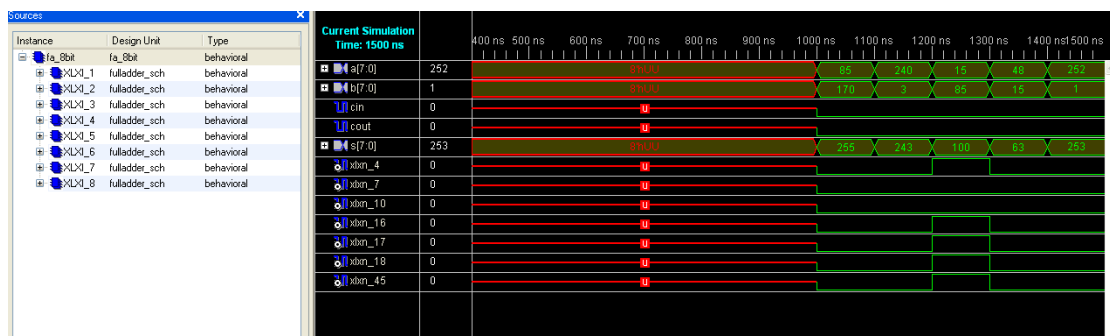
### ● 1bit adder schematic



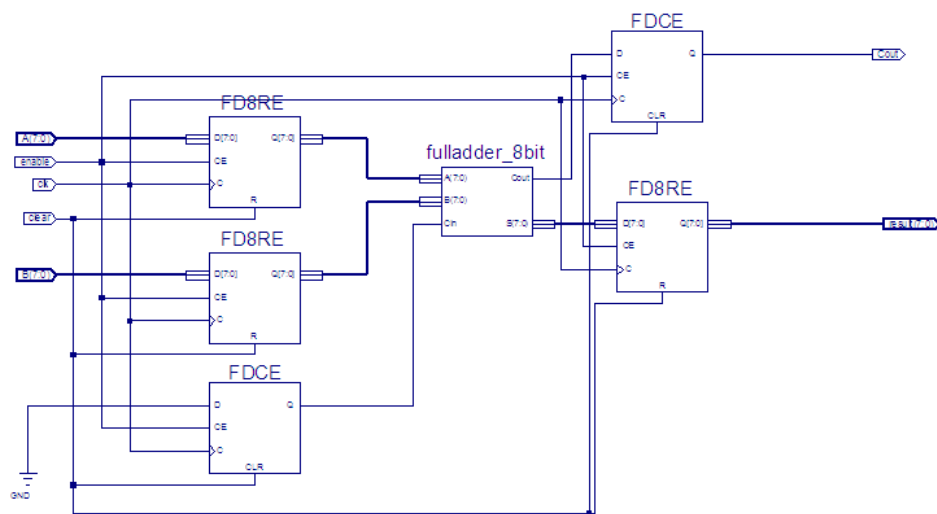
### ● 8bit adder schematic



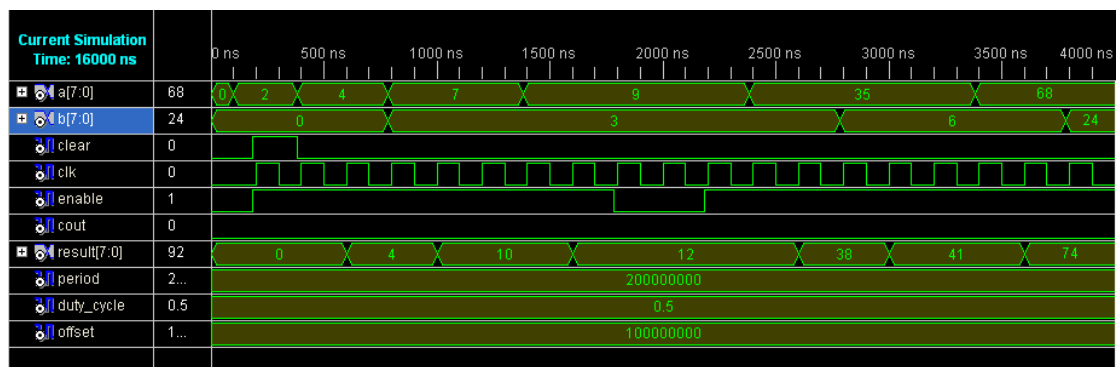
### ● 8bit adder Simulation



● 8bit adder with D Flip Flop

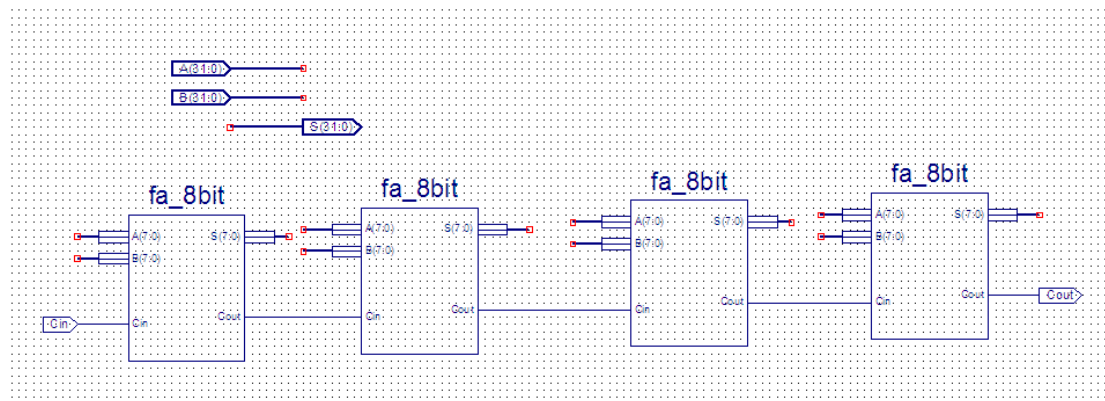


● 8bit adder with D Flip Flop Simulation

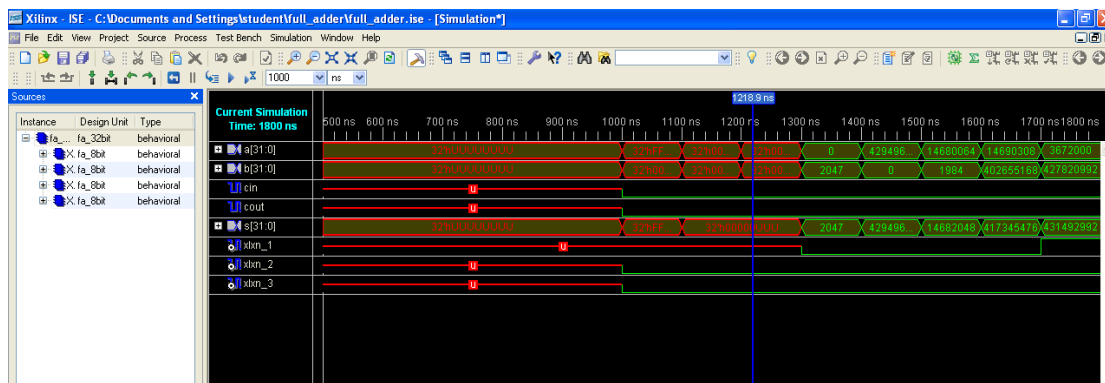


## 2. 8-bit Adder into a 32-bit Adder by instantiating and connecting 4 adders

### ● 32bit adder



### ● 32bit adder simulation

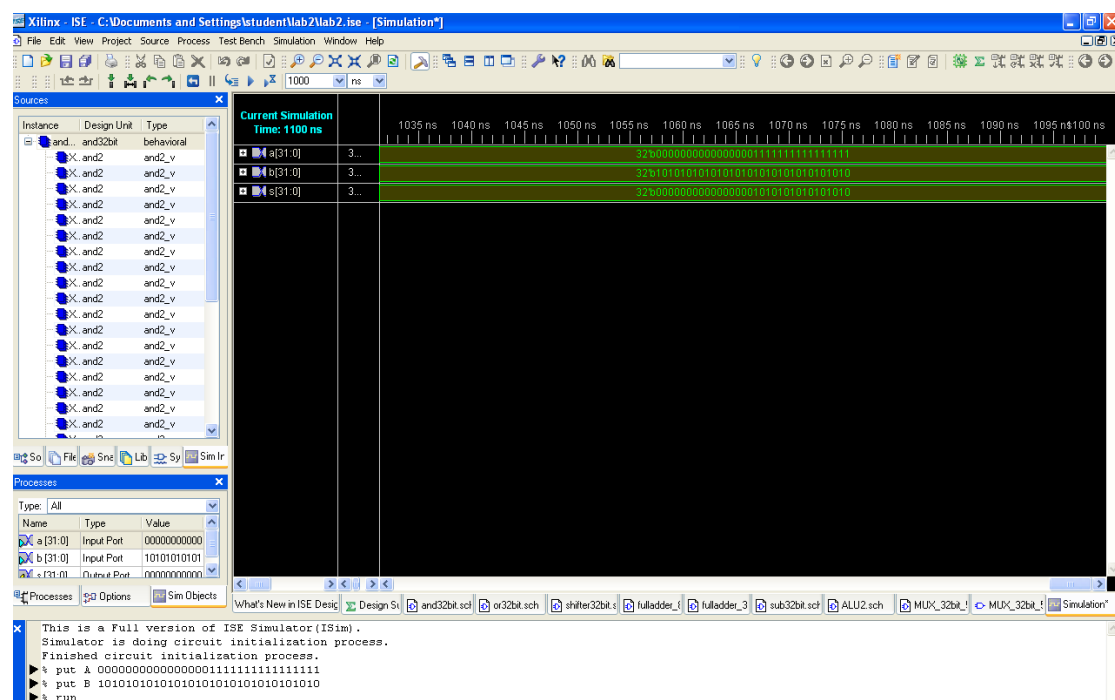


## 3. ALU

The ALU in this project supports five operations: ADD, SUB, AND, OR, and SHIFT.

Both A and B are 32-bit values. A control signal selects the operation

| SEL | Function |
|-----|----------|
| 000 | ADD      |
| 001 | SUB      |
| 010 | AND      |
| 011 | OR       |
| 1XX | Shift    |

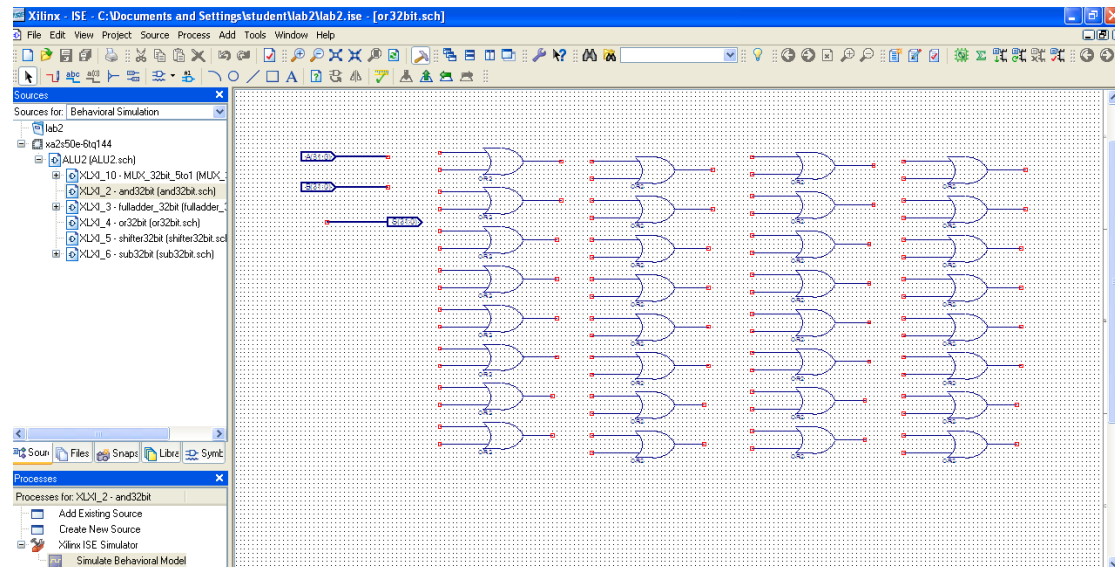


## b. OR

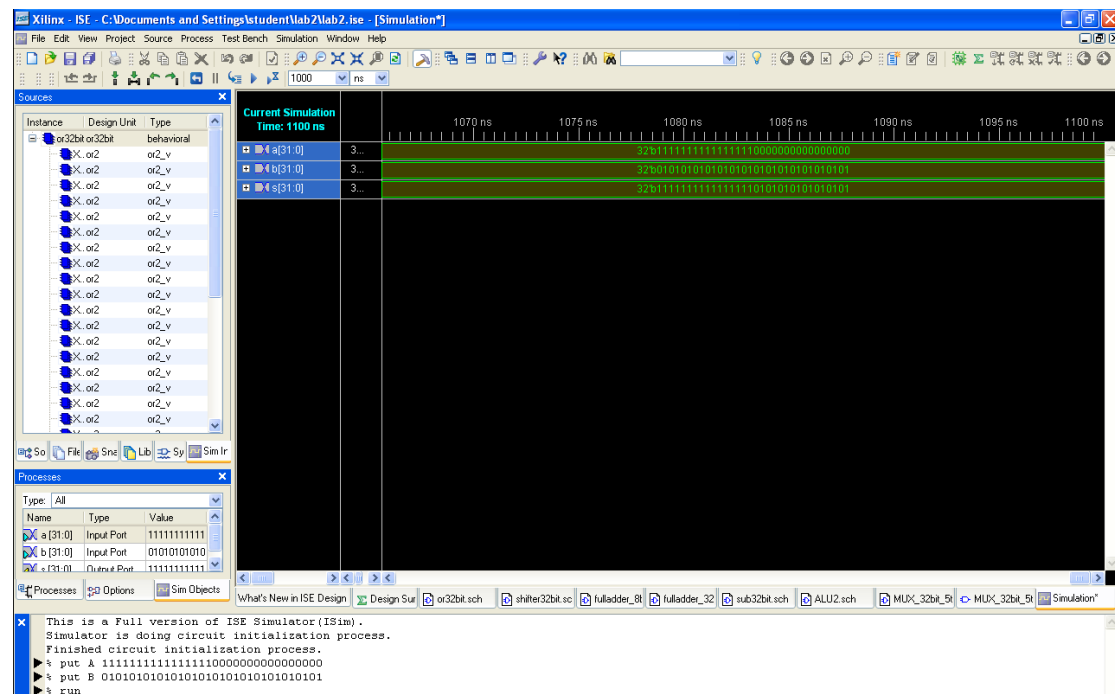
Performs a logical OR operation on each corresponding bit of the two 32-bit inputs.

$$Si = Ai + Bi$$

### ● 32 bit OR schematic



### ● 32 bit OR Simulation



### c. shifter

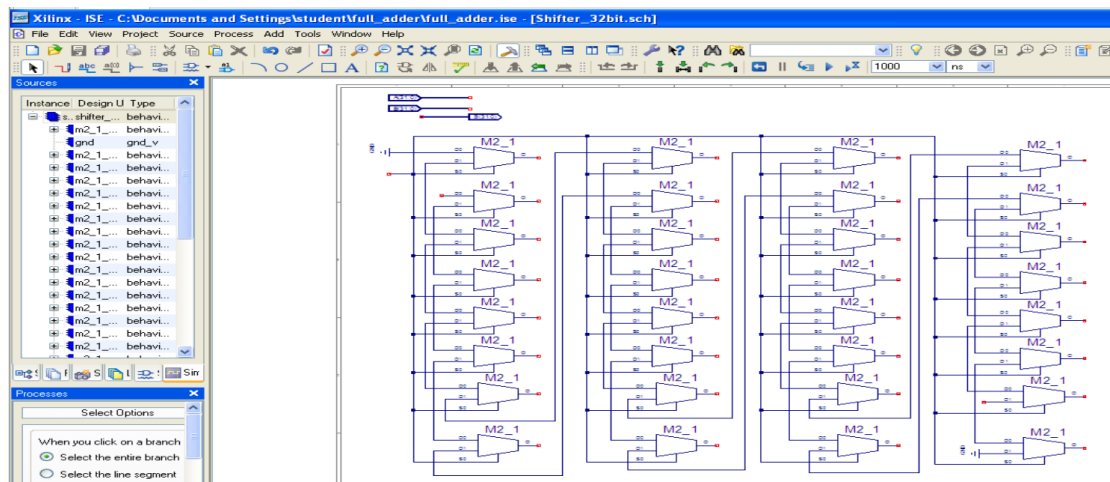
**Function:** Shifts the 32-bit input vector left or right based on a selection bit.

**Implementation Logic:** The selection bit depends on B's LSB.

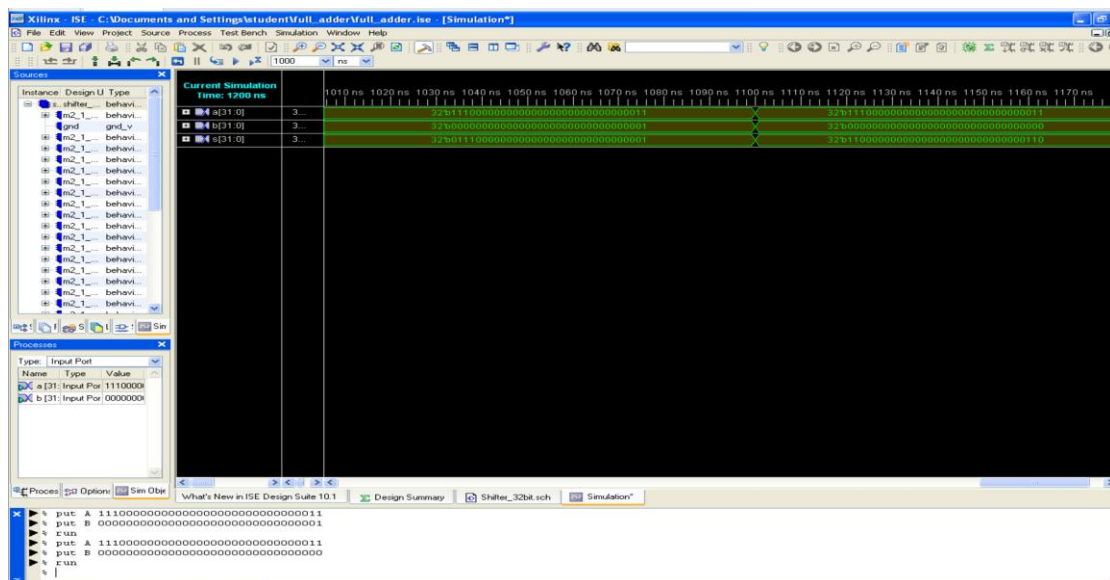
Selection 0: Executes Left Shift.

Selection 1: Executes Right Shift.

#### ● Shifter schematic



#### ● Shifter Simulation

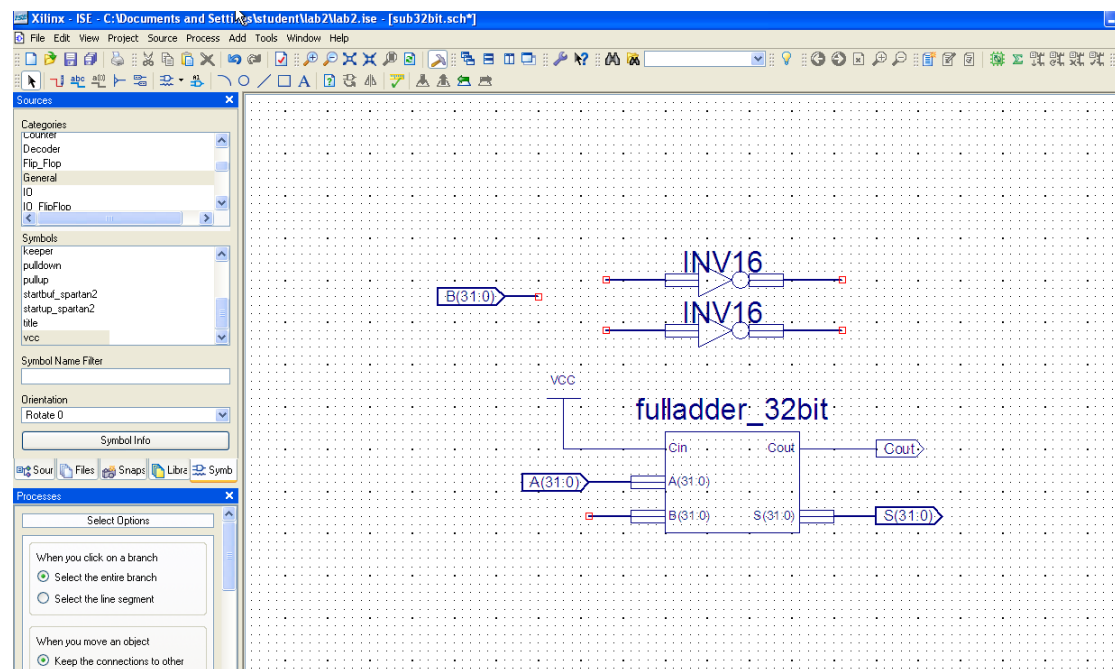


#### d. Subtractor

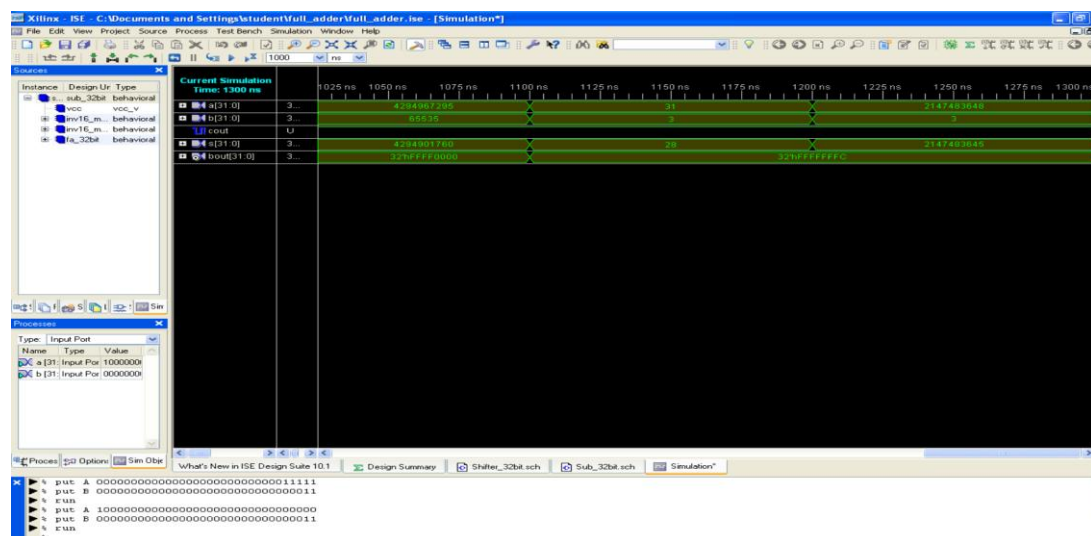
**Function:** Performs the arithmetic subtraction of two 32-bit inputs (A - B).

**Implementation:** Following the principles of 2's complement arithmetic, the bits of input B are inverted, and the initial carry-in (C\_in) of the 32-bit adder is set to 1 to complete the subtraction logic.

#### ● Subtractor schematic



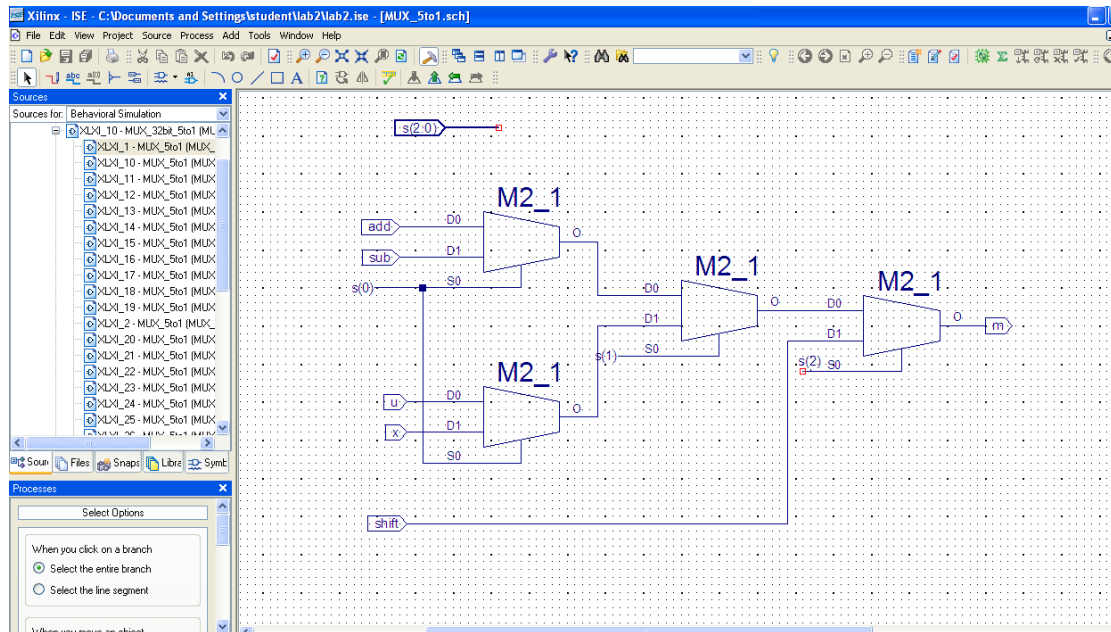
#### ● Subtractor Simulation



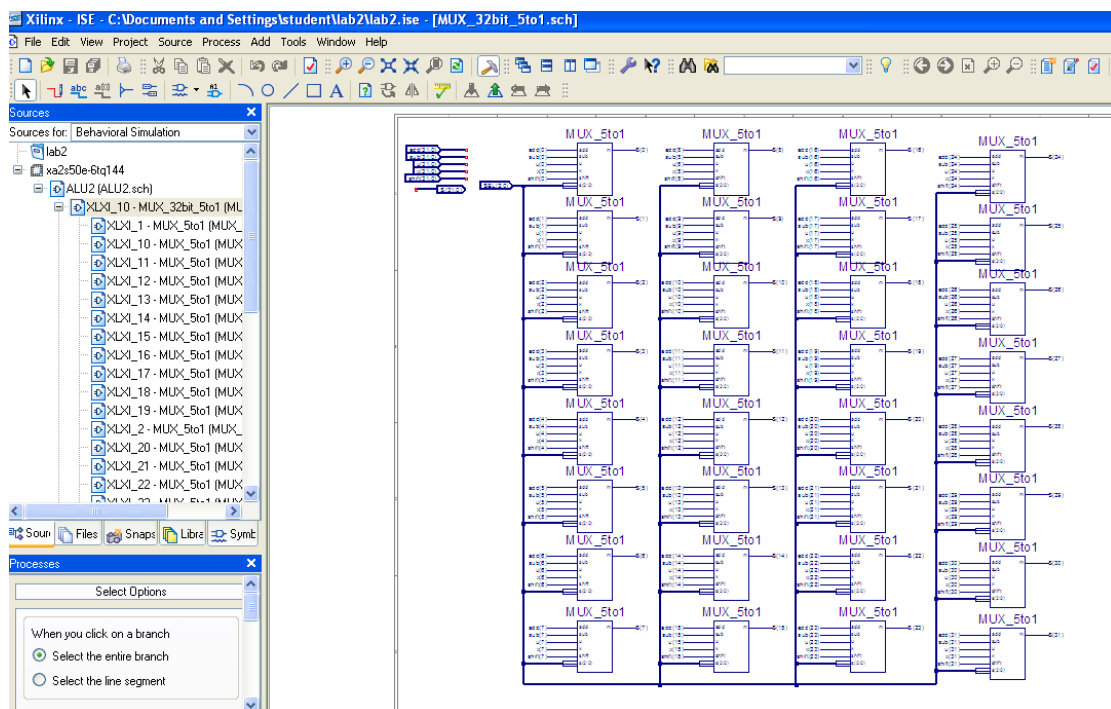
### e. 5-to-1 MUX

A 5-to-1 Multiplexer was designed to select between the results of the Adder, Subtractor, Shifter, AND, and OR operations based on the user-defined operation code.

#### ● 5-to-1 mux



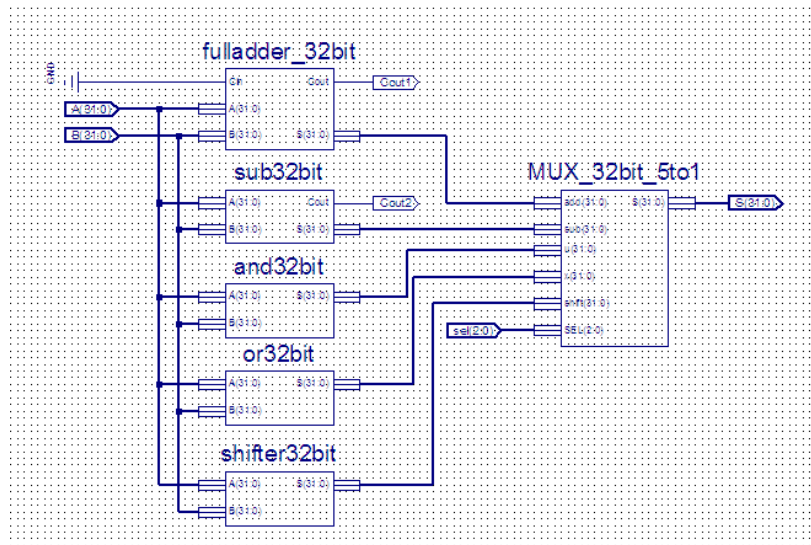
#### ● 5-to-1 mux 32bit





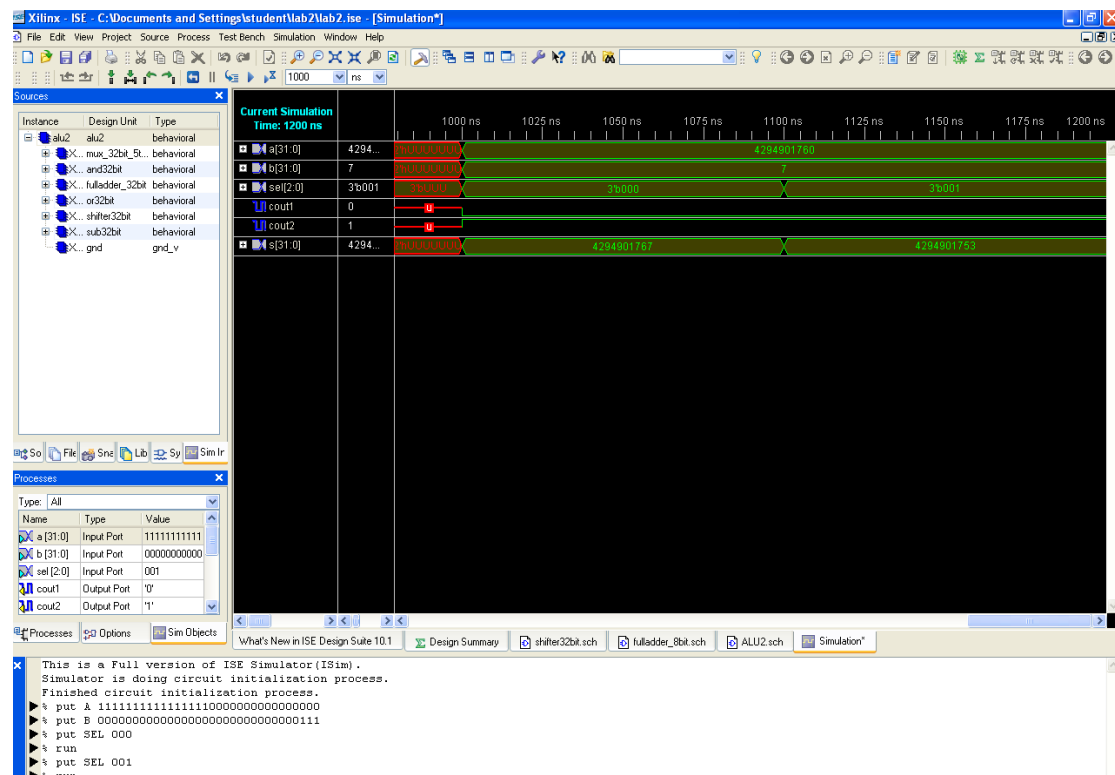
## f. ALU

### ● ALU schematic

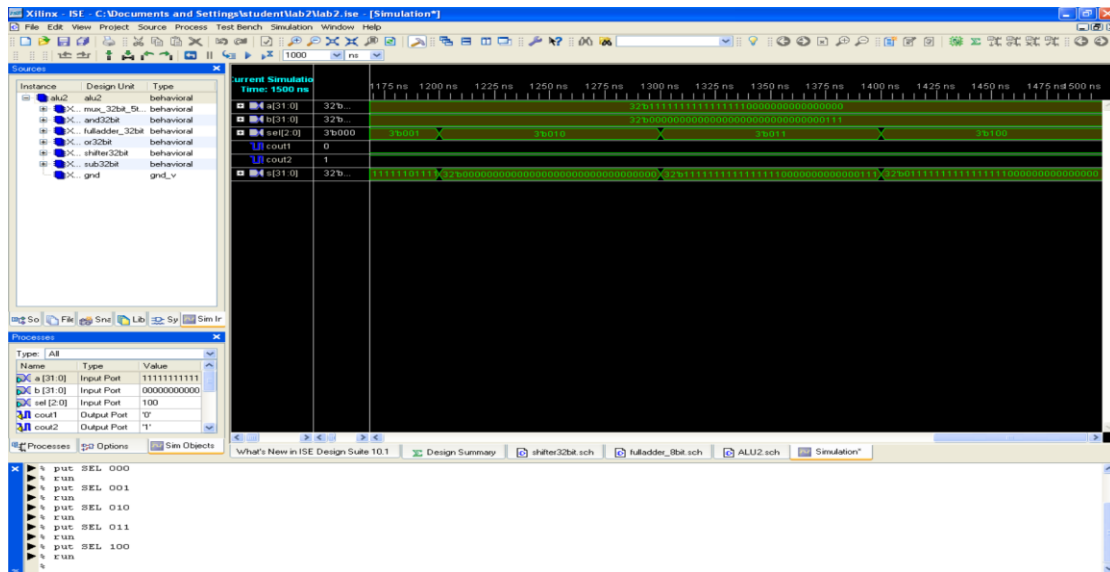


### ● ALU Simulation

#### ADD and SUB function

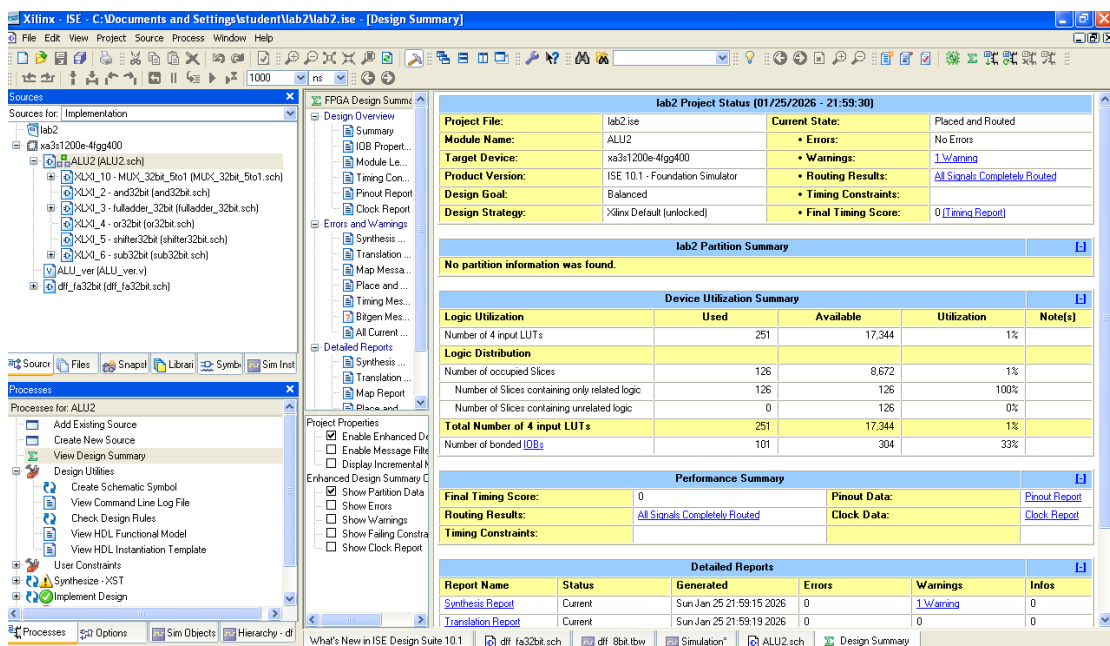


## AND / OR/ Shift function



## ● Log file

Number of 4-input LUTs = 251.



## 4. ALU Verilog

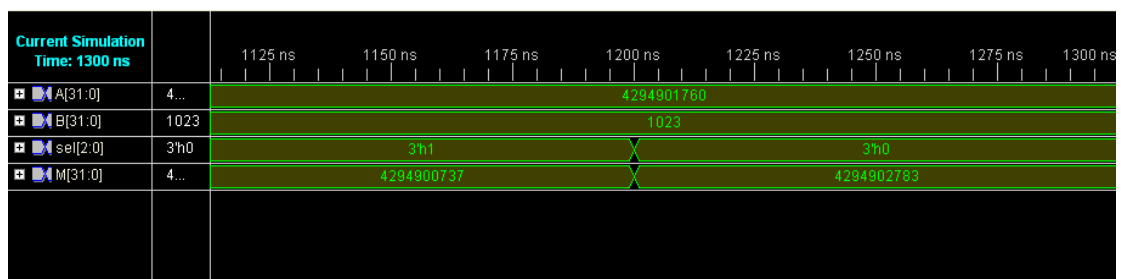
[https://github.com/rita0128/EE533\\_lab2](https://github.com/rita0128/EE533_lab2)

### ● Code

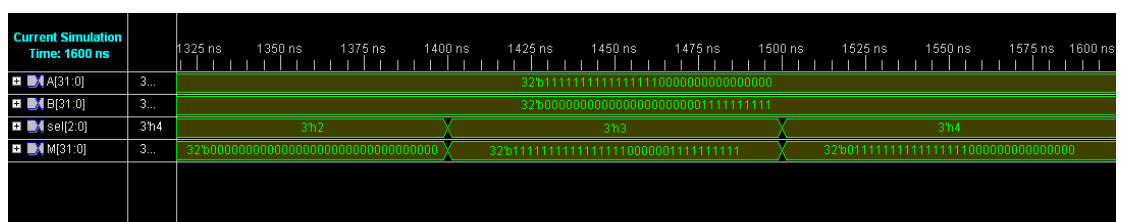
```
21 module ALU_ver(  
22     input [31:0] A,  
23     input [31:0] B,  
24     input [2:0] sel,  
25     output reg [31:0] M  
26 );  
27 always@(*)begin  
28     case(sel)  
29         3'b000: begin //ADD  
30             M = A + B;  
31         end  
32  
33         3'b001: begin //SUB  
34             M = A - B;  
35         end  
36  
37         3'b010: begin //AND  
38             M = A & B;  
39         end  
40  
41         3'b011: begin //OR  
42             M = A | B;  
43         end  
44  
45         3'b100: begin //SHIFT  
46             if (B[0] == 1'b0)  
47                 M = A << 1;  
48             else  
49                 M = A >> 1;  
50         end  
51  
52         default: begin  
53             M = 32'b0;  
54         end  
55     endcase  
56 end  
57 endmodule
```

### ● Simulation

#### ADD and SUB function



#### AND / OR / Shift function



## ● Log file

Number of 4-input LUTs = 148.

The screenshot shows the Xilinx ISE Design Summary window for project 'lab2.ise'. The summary is divided into several sections:

- Project File:** lab2.ise, Current State: Placed and Routed.
- Module Name:** ALU\_ver, Errors: No Errors.
- Target Device:** xc3s1200e-4fgg400, Warnings: No Warnings.
- Product Version:** ISE 10.1 - Foundation Simulator, Routing Results: All Signals Completely Routed.
- Design Goal:** Balanced, Timing Constraints: 0 (Timing Report).
- Design Strategy:** Xilinx Default (unlocked), Final Timing Score: 0 (Timing Report).

**lab2 Partition Summary**

No partition information was found.

**Device Utilization Summary**

| Logic Utilization                              | Used       | Available     | Utilization | Note(s) |
|------------------------------------------------|------------|---------------|-------------|---------|
| Number of 4 input LUTs                         | 148        | 17,344        | 1%          |         |
| Logic Distribution                             |            |               |             |         |
| Number of occupied Slices                      | 80         | 8,672         | 1%          |         |
| Number of Slices containing only related logic | 80         | 80            | 100%        |         |
| Number of Slices containing unrelated logic    | 0          | 80            | 0%          |         |
| <b>Total Number of 4 input LUTs</b>            | <b>148</b> | <b>17,344</b> | <b>1%</b>   |         |
| Number used as a route-thru                    | 12         |               |             |         |
| Number of bonded I/Os                          | 99         | 304           | 32%         |         |

**Performance Summary**

| Final Timing Score: | 0                             | Pinout Data: | Pinout Report |
|---------------------|-------------------------------|--------------|---------------|
| Routing Results:    | All Signals Completely Routed | Clock Data:  | Clock Report  |
| Timing Constraints: |                               |              |               |

**Detailed Reports**

| Report Name            | Status      | Generated                | Errors | Warnings | Info    |
|------------------------|-------------|--------------------------|--------|----------|---------|
| Synthesis Report       | Current     | Sun Jan 25 22:06:30 2026 | 0      | 0        | 1 Info  |
| Translation Report     | Current     | Sun Jan 25 22:06:39 2026 | 0      | 0        | 0       |
| Map Report             | Current     | Sun Jan 25 22:06:42 2026 | 0      | 0        | 1 Info  |
| Place and Route Report | Current     | Sun Jan 25 22:06:46 2026 | 0      | 0        | 1 Info  |
| Static Timing Report   | Current     | Sun Jan 25 22:06:49 2026 | 0      | 0        | 3 Infos |
| Bitstream Report       | Out of Date | Sat Jan 24 21:37:09 2026 | 0      | 0        | 0       |